

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	SRIKRISHNA.P	Reg. No.:	192511435
Section / Batch:	2025-2029	Date:	08/07/2026

Code of Conduct

I SRIKRISHNA P / 192511435 (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: SRIKRISHNA.P

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1
C Code - Pointer with Array
10 marks**Q2**
C Code - Linked List Node
10 marks**Q3**
C Code - Binary Search
Complexity
10 marks**Q4**
C Code - Pointer Increment
10 marks**Q5**
C Code - Array Address
10 marks**Q6**
C Code - Linked List Traversal
10 marks**Q7**
C Code - Recursion (Factorial)**Q1 C Code - Pointer with Array**

10 marks

```
#include<stdio.h>
int main(){
    int arr[]={10,20,30};
    int *p=arr;
    printf("%d",*(p+2));
}
```

Your Answer

30

[Save Answer](#)[Next →](#)

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1
C Code - Pointer with Array
10 marks**Q2**
C Code - Linked List Node
10 marks**Q3**
C Code - Binary Search
Complexity
10 marks**Q4**
C Code - Pointer Increment
10 marks**Q5**
C Code - Array Address
10 marks**Q6**
C Code - Linked List Traversal
10 marks**Q2 C Code - Linked List Node**

10 marks

```
#include<stdio.h>
struct Node{
    int data;
    struct Node *next;
};
int main(){
    struct Node n1,n2;
    n1.data=10;
    n2.data=20;
    n1.next=&n2;
    n2.next=NULL;
    printf("%d",n1.next->data);
}
```

Your Answer

20

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

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C Code - Pointer with Array
10 marks**Q2**
C Code - Linked List Node
10 marks**Q3**
C Code - Binary Search
Complexity
10 marks**Q4**
C Code - Pointer Increment
10 marks**Q5**
C Code - Array Address
10 marks**Q6**
C Code - Linked List Traversal
10 marks**Q7**
C Code - Recursion (Factorial)**Q3 C Code - Binary Search Complexity**

10 marks

```
while(low<=high){
    mid=(low+high)/2;
}
```

Your Answer

Time Complexity: O(log n)
Space Complexity: O(1)

[Save Answer](#)[Next →](#)

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

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C Code – Binary Search
Complexity
10 marks**Q4**
C Code – Pointer Increment
10 marks**Q5**
C Code – Array Address
10 marks**Q6**
C Code – Linked List Traversal
10 marks**Q7**
C Code – Recursion (Factorial)**Q4 C Code - Pointer Increment**

10 marks

```
#include<stdio.h>

int main(){
    int a[]={5,10,15};
    int *p=a;
    p++;
    printf("%d",*p);
}
```

Your Answer

10

[Save Answer](#)[Next →](#)

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1
C Code – Pointer with Array
10 marks**Q2**
C Code – Linked List Node
10 marks**Q3**
C Code – Binary Search
Complexity
10 marks**Q4**
C Code – Pointer Increment
10 marks**Q5**
C Code – Array Address
10 marks**Q6**
C Code – Linked List Traversal
10 marks**Q7**
C Code – Recursion (Factorial)**Q5 C Code - Array Address**

10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

Your Answer

10

[Save Answer](#)[Next →](#)

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1
C Code – Pointer with Array
10 marks**Q2**
C Code – Linked List Node
10 marks**Q3**
C Code – Binary Search
Complexity
10 marks**Q4**
C Code – Pointer Increment
10 marks**Q5**
C Code – Array Address
10 marks**Q6 C Code – Linked List Traversal**

10 marks

```
Node1 -> Node2 -> Node3 (Node values are 10,20,30)
current=head;
current=current->next;
printf("%d",current->data);
```

Your Answer

20

[Save Answer](#)[Next →](#)

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1

C Code – Pointer with Array
10 marks

Q2

C Code – Linked List Node
10 marks

Q3

C Code – Binary Search
Complexity
10 marks

Q4

C Code – Pointer Increment
10 marks

Q5

C Code – Array Address
10 marks

Q6



Q7 C Code – Recursion (Factorial)

10 marks

```
#include <stdio.h>
int fact(int n){
    if(n==0)
        return 1;
    return n * fact(n-1);
}
int main(){
    printf("%d", fact(4));
    return 0;
}
```

Your Answer

24

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1

C Code – Pointer with Array
10 marks

Q2

C Code – Linked List Node
10 marks

Q3

C Code – Binary Search
Complexity
10 marks

Q4

C Code – Pointer Increment
10 marks

Q5

C Code – Array Address
10 marks

Q6

C Code – Linked List Traversal
10 marks

Q8 C Code – Recursion (Sum)

10 marks

```
#include <stdio.h>
int sum(int n){
    if(n==0)
        return 0;
    return n + sum(n-1);
}
int main(){
    printf("%d", sum(5));
}
```

Your Answer

15

CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1

C Code – Pointer with Array
10 marks

Q2

C Code – Linked List Node
10 marks

Q3

C Code – Binary Search
Complexity
10 marks

Q4



Q9 C Code – Primitive Data Type

10 marks

```
#include <stdio.h>
int main(){
    int a=10;
    char ch='A';
    printf("%d %c",a,ch);
}
```

Your Answer

10 A

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CO1_AT1_C CODE OUTPUT PREDICTION

[← Back](#)

QUESTIONS

Q1

C Code – Pointer with Array
10 marks

Q2

C Code – Linked List Node
10 marks

Q3

C Code – Binary Search
Complexity
10 marks

Q4

C Code – Pointer Increment
10 marks

Q5 C Code – Linear Data Structure (Array)

10 marks

```
#include <stdio.h>
int main(){
    int arr[]={2,4,6,8};
    printf("%d",arr[3]);
}
```

Your Answer

8